Juneau Water Distribution Network

Synthetic GIS infrastructure model, spatial database, network tracing and web mapping | Technical portfolio case study | October 2026

1. Executive summary

This independent project demonstrates end-to-end GIS development for a simulated municipal water-distribution system in Juneau, Alaska. Public road centerlines, parcel geometry and hydrant locations provided geographic reference inputs. ArcPy generated synthetic water mains, service connections and devices; ArcGIS Pro and PostgreSQL/PostGIS supported spatial processing and topology-oriented database work; and ArcGIS-hosted feature layers plus a JavaScript web map provide interactive network exploration.

Critical qualification: All modeled water infrastructure, connections, facilities, attributes and flow relationships are synthetic. Real reference features do not establish the existence, ownership, engineering characteristics, hydraulic performance or operating status of any actual Juneau water asset.

2. Objectives and deliverables

The project was designed to demonstrate automated GIS feature generation; geodatabase schema design; coordinate-system management; data-quality checks; SQL/PostGIS spatial and graph-oriented analysis; utility isolation concepts; and delivery of an interactive public-facing GIS demonstration. Deliverables include generated geodatabase feature classes, staged network tables and functions, published feature services, and a static web dashboard hosted using GitHub Pages.

3. Geographic reference data and provenance

Reference dataset	Role in model	Provenance status
Road centerlines	Candidate synthetic main alignments	City and Borough of Juneau provides road centerline downloads; exact acquired file/version not independently verified
Fire hydrant locations	Spatial anchors for candidate mains and synthetic hydrants	City GIS public hydrant service identified; exact source snapshot not independently verified
Parcel polygons	Potential customer locations and service candidates	City parcel downloads and public viewer identified; exact acquired file/version not independently verified

Input inventory recorded during development: **1,983 road centerlines; 1,470 hydrants; 16,250 parcels**. These are working-project counts, not official agency totals. A source-data manifest with acquisition dates, item IDs, license terms and file checksums would strengthen reproducibility.

4. Projection and geodatabase design

The initial project used NAD 1983 (2011) StatePlane Alaska 1, US survey feet, for local distance-based generation. During database staging, a spatial-reference identifier compatibility issue required a revised import/projection approach; EPSG:6393 was considered for PostGIS staging. A reproducible build should record the exact coordinate system and unit of every final table, and validate transformations before spatial joins or distance calculations.

The ArcPy generator wrote a separate file geodatabase and preserved the source inputs. Principal feature classes included WaterMain, MainJunction, Hydrant, Valve, Meter, ServiceJunction, ServiceLine, CustomerParcelPoint and QA_UnservedParcels. The generator itself did not create a licensed Esri Utility Network.

5. Synthetic network generation

The documented Python/ArcPy generator selected road segments within 500 feet of public hydrant locations as candidate mains. Hydrants were snapped to generated mains only within 150 feet; parcels were eligible for modeled service connections when within 250 feet of a candidate main. Synthetic valves were distributed at approximately 1,200-foot intervals and at a

randomized 18% of distinct main junctions. The generator used random seed 42 and simulated installation years from 1950 to 2024. These are modeling parameters, not measured infrastructure facts.

Generated feature or QA category	Recorded project count
Candidate mains	1,910
Main junctions	1,678
Hydrants snapped	1,376 of 1,470 reference hydrants
Hydrant laterals	1,413
Service candidates / meters	13,192
Valves	606
QA: unserved parcels	3,057

Counts reflect recorded intermediate output and may not match the later directed web graph or expanded asset-package dataset. In particular, the 3,057 unserved-parcel QA count should not be interpreted as a measured deficiency in municipal service.

6. Utility Network exploration and database migration

An expanded ArcGIS Utility Network-oriented asset package was assembled with WaterLine (23,628), WaterJunction (10,905) and WaterDevice (20,836) features. Attempts to load or configure the Utility Network encountered licensing and ObjectID/field-editing constraints. The project consequently shifted toward PostgreSQL/PostGIS and QGIS for open, inspectable network analysis rather than claiming a fully operational Esri Utility Network.

PostGIS work used separate staging and analysis-oriented structures. Troubleshooting included duplicate hydrant/node identifiers, geometry/SRID consistency, edge-node relationships and feature publication requirements. An ArcGIS Online publication analyzer flagged 64-bit ObjectIDs; subsequent publishing work required verifying 32-bit compatible ObjectID fields. The presence of a feature class alone does not prove connected topology.

7. Graph and valve-isolation analysis

The database work modeled water mains as graph edges connected through junction nodes, with synthetic valves associated with edge endpoints. SQL/PostGIS functions were developed to inspect edge isolation readiness and identify valves associated with candidate main segments. One recorded test edge, ID 2199 (main WM0001885), had from-node 1383, to-node 1384, two endpoint valves (butterfly and gate), and isolation_ready=true. This validates the test record only; it does not establish network-wide isolation completeness.

A later web tracing model used directed edges and precomputed reachability. Four synthetic source groups replaced an earlier six-source concept. Source-group counts recorded during rebuilding were A 879, B 604, C 1,454 and D 622; these are source-associated counts from a particular processing stage, not necessarily unique pipe totals. The published HTML explicitly describes its direction as graph-based rather than hydraulically simulated.

8. Interactive GIS dashboard and publication

The web mapping implementation uses the ArcGIS Maps SDK for JavaScript 4.34, an Esri topographic basemap, an ArcGIS Online hosted mains FeatureLayer, and browser-side highlighting. One retrieved dashboard version supports clicking a main and tracing upstream/downstream connectivity. Its graph files include juneau_directed_edges_v2.csv, juneau_node_index_v2.json and juneau_reachability_v2.bin; the JavaScript expects 1,995 indexed nodes. The website is delivered as static HTML/JavaScript and supporting assets through GitHub Pages, while geographic features are loaded from ArcGIS-hosted services.

A separate valve-closure/isolation visualization was developed during the portfolio work to illustrate affected mains, services and parcels. This capability should be presented as a synthetic scenario analysis, not as an operational outage predictor. The

precise live GitHub Pages URL and currently deployed feature set should be inserted after checking the production site.

Published feature service references:
Mains: https://services8.arcgis.com/B0RimbuxBeyw78KX/arcgis/rest/services/Juneau_Water_Network_Mains/FeatureServer/19
Legacy supply points (hidden in retrieved v2 dashboard):
https://services8.arcgis.com/B0RimbuxBeyw78KX/arcgis/rest/services/Juneau_Supply_Points/FeatureServer/1

9. Quality assurance, known constraints and interpretation

QA covered unsnapped hydrants, unserved parcel candidates, node/edge identifiers, topology consistency, coordinate reference systems and publishing schema compatibility. Limitations include synthetic road-following main placement; no field survey or as-built verification; incomplete representation of real valves, pressure zones and controls; no calibrated hydraulic solver; simplified directed graph assumptions; and the possibility that some gray/unassigned mains lie outside the directed web trace model. Reachability and isolation results are analytical demonstrations, not engineering decisions.

10. Recommended reproducibility and future improvements

For a fully reproducible release, preserve a dated source manifest; publish versioned generator scripts and configuration; document table schemas and field definitions; add automated checks for orphan nodes, duplicated IDs and disconnected components; store explicit build counts at each stage; verify that deployed HTML and binary graph assets are synchronized; and document differences between upstream/downstream trace and valve-closure scenarios. Future extensions could incorporate pgRouting or a hydraulic engine, but neither should be implied to have been completed.

11. Verified public data and software references

City and Borough of Juneau - Maps and GIS
https://juneau.org/community-development/maps

City and Borough of Juneau - Parcel Viewer and downloadable parcel/road centerlines
https://epv.ci.juneau.ak.us/

City and Borough of Juneau - base map service (includes fire_hydrants layer 4)
https://epv.ci.juneau.ak.us/arcgis/rest/services/1234_CBJ_Base_Map/MapServer

Alaska DNR - archived borough parcel feature service (not necessarily project source)
https://arcgis.dnr.alaska.gov/arcgis/rest/services/OpenData/Administrative_BoroughParcels/FeatureServer/3

ArcGIS Maps SDK for JavaScript
https://developers.arcgis.com/javascript/latest/

PostGIS documentation
https://postgis.net/documentation/

GitHub Pages documentation
https://docs.github.com/en/pages

Source attribution note: These URLs identify relevant public providers/services and software documentation. They do not prove that the precise current hosted records are identical to the datasets originally downloaded for the synthetic model.